

A Point-Like Dark-Matter Target at 12.8 GHz from a QCD-Anchored Vacuum Condensate: Theory and Haloscope Test Protocol

Oleg Kirichenko

Independent Researcher; urevich55@gmail.com
<https://github.com/infosave2007/vmf>

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Abstract

The Null-Vector Gravity (NVG) framework, anchored to a single QCD input $M_{\Omega,0} = 859 \pm 8$ MeV fixed by lattice sigma terms, contains a pseudo-Goldstone mode θ of the vacuum condensate $\Phi = W e^{i\theta}$. We examine the phenomenological consequences of the hypothesis that the same decay constant f_a that sets the θ -mode mass also generates Majorana neutrino masses through a Weinberg-like dimension-five operator suppressed by a two-loop QCD-anomaly coefficient. Introducing the unknown overall coefficient of this operator as an explicit parameter c_ν , the benchmark choice $c_\nu = 1$ fixes $f_a = 1.07 \times 10^{11}$ GeV from the neutrino sector and yields a *conditional point target* for dark matter: a coherent pseudo-scalar of mass $m_\theta = 53.2$ μ eV, detectable as a microwave line at $f_\theta = 12.86$ GHz with photon coupling $g_{\theta\gamma\gamma} = 2.08 \times 10^{-14}$ GeV $^{-1}$ (KSVZ benchmark). The product $m_\theta f_a$ agrees with the QCD axion band at 2.6% — a normalization cross-check, not an independent test — and the benchmark frequency lies in the currently unexplored gap between the published ORGAN mass windows (25–26 and 63–67 μ eV), directly on the lower edge of the SMASH band (50–200 μ eV) that ORGAN has declared as its roadmap target. We present the derivation, a calibration of the haloscope signal formula against the published ORGAN signal estimate, a Monte Carlo propagation of the statistical input uncertainties (m_3, χ_{top}), together with explicit conditional sensitivity scenarios for the unknown coefficient c_ν , and two concrete test protocols: (A) a resonant-cavity array and (B) a *fixed-position* dielectric booster stack, where the point target eliminates the scanning mechanism. For a coherent dark-matter fraction of 24% (misalignment only), protocol B reaches SNR = 5 in ~ 18 h at power boost $\beta^2 = 10^4$ (MADMAX convention, $\beta^2 \equiv P_{\text{sig}}/P_0$). Discovery and exclusion criteria are preregistered. This is a *phenomenological target paper*: the θ -seesaw operator is stated as a hypothesis, its coefficient is propagated as a parameter, and every derived quantity is conditional on it. All numerical values are reproducible from the accompanying scripts.

Keywords: axion dark matter, haloscope, dielectric haloscope, pseudo-Goldstone boson, vacuum condensate, neutrino mass, phenomenological target

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1 Introduction

Direct searches for QCD axion dark matter have so far produced exclusion limits in disjoint mass windows: ADMX covers $\sim 2.7\text{--}4.2\ \mu\text{eV}$ [14], HAYSTAC covers $\sim 17\text{--}20\ \mu\text{eV}$ [15], ORGAN has scanned $25.45\text{--}26.27\ \mu\text{eV}$ [18], $63.2\text{--}67.1\ \mu\text{eV}$ [16] and $107.42\text{--}111.93\ \mu\text{eV}$ [17], while MADMAX is under construction for the $40\text{--}400\ \mu\text{eV}$ range [21]. The intermediate region around $50\text{--}60\ \mu\text{eV}$ ($12\text{--}15\ \text{GHz}$) remains unexplored.

This note reports that the NVG/VMF framework [2], anchored to a single QCD input [1], yields — conditional on one stated hypothesis (the θ -seesaw, Sec. 2.3) — a point target *exactly* in that gap: a dark-matter pseudo-scalar at $53.2\ \mu\text{eV}$. Because the target is a *point* in (m, g_γ) space rather than a band, the experimental problem simplifies drastically: no scanning, no tuning mechanism — a fixed-frequency integration.

Section 2 gives the derivation. Section 3 states the observable signature. Section 4 documents the gap in published searches. Section 5 defines the test protocol, including a calibration of the signal formula against published data and preregistered discovery/exclusion criteria. Section 6 lists the framework assumptions honestly.

2 Theoretical Framework

2.1 The anchor: $M_{\Omega,0}$ from lattice sigma terms

The framework is anchored to one dimensionful QCD input. Separating the nucleon mass into current-quark and nonperturbative components via the Feynman–Hellmann sigma terms ($\sigma_{\pi N} \approx 44\ \text{MeV}$, $\sigma_{sN} \approx 30\ \text{MeV}$), the vacuum value of the nonperturbative component is fixed at [1]

$$M_{\Omega,0} = 859 \pm 8\ \text{MeV}, \tag{1}$$

which sets the condensate energy density scale $\rho_c \approx 7.09 \times 10^4\ \text{MeV}/\text{fm}^3$ used throughout the construction. No cosmological or astrophysical input is used at any stage.

2.2 The θ pseudo-Goldstone mode

The NVG vacuum condensate is parametrized as $\Phi = W e^{i\theta}$, where the phase θ is a pseudo-Goldstone mode with the standard anomaly-generated potential. Its mass obeys the same

relation as the QCD axion,

$$m_\theta = \frac{\sqrt{\chi_{\text{top}}}}{f_a}, \quad (2)$$

with the topological susceptibility $\chi_{\text{top}}^{1/4} = 75.5$ MeV from lattice QCD [10]. In the normalization $m_a(f_a = 10^{12} \text{ GeV}) = 5.691 \text{ } \mu\text{eV}$ [10], this becomes $m_\theta = 5.691 \text{ } \mu\text{eV} \times (10^{12} \text{ GeV}/f_a)$.

2.3 The θ -seesaw: fixing f_a from the neutrino sector

The framework hypothesis behind the target is the θ -seesaw: the same mode θ generates Majorana neutrino masses without right-handed neutrinos. To be consistent with the electroweak gauge group above EWSB, the operator must have the unique gauge-invariant dimension-five (Weinberg) structure [7]; the NVG hypothesis specifies only its suppression scale and coefficient,

$$\mathcal{L}_{\text{eff}} = 2 c_\nu \left(\frac{\alpha_s}{4\pi} \right)^2 \frac{1}{f_a} (\tilde{H}^\dagger L)^T C (\tilde{H}^\dagger L) + \text{h.c.}, \quad \tilde{H} \equiv i\sigma_2 H^*, \quad (3)$$

where c_ν is an unknown dimensionless coefficient. Since c_ν already absorbs the UV normalization, we *define* it through the physical relation (4) below; the overall prefactor of Eq. (3) — including the displayed factor of 2, which compensates the two factors of $1/\sqrt{2}$ from $\langle H \rangle = v_{\text{EW}}/\sqrt{2}$ — is then conventional and carries no independent physical meaning. After EWSB the operator reduces to the Majorana mass, for the heaviest neutrino,

$$m_3 = c_\nu \left(\frac{\alpha_s}{4\pi} \right)^2 \frac{v_{\text{EW}}^2}{f_a}. \quad (4)$$

With $\alpha_s(M_Z) = 0.1184$, $v_{\text{EW}} = 246.22$ GeV and the oscillation-and-cosmology anchor $m_3 = 50.3$ meV (normal ordering; the implied sum $\sum m_\nu = 59$ meV passes the DESI DR2 bound $\sum m_\nu < 64$ meV unconditionally), Eq. (4) fixes

$$f_a = c_\nu \left(\frac{\alpha_s}{4\pi} \right)^2 \frac{v_{\text{EW}}^2}{m_3} = 1.07 \times 10^{11} c_\nu \text{ GeV} \quad (5)$$

We stress the status of Eq. (3): it is a *framework hypothesis*, the single non-standard ingredient of this note. The nominal coefficient $(\alpha_s/4\pi)^2 \approx 8.9 \times 10^{-5}$ is motivated by the two-loop QCD-anomaly diagram, but possible flavor factors and RG running are not computed here, and the UV completion that generates the operator is not derived. The parameter c_ν absorbs all of this freedom: the benchmark $c_\nu = 1$ defines the headline target, and every result below is rescaled with c_ν (Section 5.4). The haloscope test discriminates the hypothesis directly (Section 6).

2.4 Point target for the dark-matter mass

Because every derived quantity depends on c_ν through f_a alone, the parametric structure is transparent (for fixed E/N):

$$f_a = f_{a,0} c_\nu, \quad m_\theta = \frac{m_{\theta,0}}{c_\nu}, \quad f_\theta = \frac{f_{\theta,0}}{c_\nu}, \quad g_{\theta\gamma\gamma} = \frac{g_{\theta\gamma\gamma,0}}{c_\nu}, \quad (6)$$

where the subscript 0 denotes the benchmark values at $c_\nu = 1$. A larger c_ν means a larger f_a , hence a *smaller* mass and frequency and a weaker photon coupling. Inserting (5) into (2):

$$m_\theta = 5.691 \text{ } \mu\text{eV} \times \frac{10^{12}}{1.07 \times 10^{11} c_\nu} = \frac{53.2}{c_\nu} \text{ } \mu\text{eV} \quad (7)$$

or, equivalently, a Compton frequency

$$f_\theta = \frac{m_\theta c^2}{h} = \frac{12.86}{c_\nu} \text{ GHz}. \quad (8)$$

The benchmark $c_\nu = 1$ gives $m_\theta = 53.2 \text{ } \mu\text{eV}$, $f_\theta = 12.86 \text{ GHz}$ throughout the rest of this note.

Point versus band. If a UV completion fixes c_ν , the target is a genuine *conditional point* in (m, g_γ, f) space. If c_ν remains free, the point degrades to a *broad conditional search band* along the one-parameter curve (6) — 4.3–38.6 GHz for $c_\nu \in [1/3, 3]$ (Sec. 5.4) — and the fixed-frequency advantage is retained only conditionally, at a chosen benchmark value of c_ν . We use the terminology “conditional point target” and “broad conditional search band” accordingly.

Normalization cross-check (not an independent test). Once Eq. (2) is accepted, the product $m_\theta f_a = \sqrt{\chi_{\text{top}}}$ is fixed by the same QCD input, so its landing on the QCD axion band $m_a f_a \simeq m_\pi f_\pi \sqrt{z}/(1+z) \approx 5.82 \times 10^{15} \text{ eV}^2$ [10] cannot count as an independent confirmation of NVG or of the seesaw hypothesis. The framework gives $m_\theta f_a = 5.67 \times 10^{15} \text{ eV}^2$: the 2.6% deviation is a numerical consistency check of two lattice normalizations (the χ_{top} of [10] against the VMF anchor), and nothing more. The genuinely independent content of the target is the *value* of f_a itself, fixed by the neutrino sector.

2.5 Coupling to photons

The effective interaction is the standard axion-electrodynamics term [8]

$$\mathcal{L} \supset -\frac{g_{\theta\gamma\gamma}}{4} \theta F_{\mu\nu} \tilde{F}^{\mu\nu} = g_{\theta\gamma\gamma} \theta \mathbf{E} \cdot \mathbf{B}, \quad g_{\theta\gamma\gamma} = \frac{\alpha}{2\pi} \frac{|E/N - 1.92|}{f_a}. \quad (9)$$

For the KSVZ benchmark [5] ($E/N = 0$) and the DFSZ benchmark [6] ($E/N = 8/3$):

$$g_{\theta\gamma\gamma}^{\text{KSVZ}} = 2.08 \times 10^{-14} \text{ GeV}^{-1}, \quad g_{\theta\gamma\gamma}^{\text{DFSZ}} = 8.10 \times 10^{-15} \text{ GeV}^{-1}. \quad (10)$$

The model-dependent factor $|E/N - 1.92|$ is the only residual theory uncertainty on the coupling. We emphasize that $g_{\theta\gamma\gamma}$ is *not predicted* by NVG: the framework fixes f_a but not the UV anomaly ratio E/N , so KSVZ and DFSZ are benchmarks, not consequences of the model. A null result at KSVZ level therefore excludes only KSVZ-like realizations (Section 5.4). The KSVZ value is used as the discovery benchmark.

2.6 Relic abundance: the honest coherent fraction

For a post-inflationary scenario with the natural initial misalignment $\theta_i = \pi/\sqrt{3}$, the standard misalignment estimate [9] gives

$$\Omega_\theta h^2 \simeq 0.12 \theta_i^2 \left(\frac{f_a}{10^{12} \text{ GeV}} \right)^{7/6} = 0.0291, \quad (11)$$

i.e. **24% of the local dark-matter density** ($\Omega_{\text{DM}} h^2 = 0.120$). The remaining density may reside in topological defects of the condensate sector; it does not contribute to the coherent haloscope signal. All signal powers below use the conservative coherent fraction $f_{\text{coh}} = 0.24$; the $f_{\text{coh}} = 1$ values are also quoted where relevant. A larger coherent fraction (e.g. from defect decay) would strengthen the signal proportionally.

2.7 Fifth force: the pseudoscalar null channel

The Compton range $\lambda_\theta = \hbar c/m_\theta = 3.72 \text{ mm}$ falls inside the sub-millimetre window of torsion-balance experiments [12]. Because θ is a *pseudoscalar*, the leading-order monopole–monopole Yukawa exchange between unmixed states vanishes, so unpolarized Eöt-Wash-class searches are expected to see nothing at this range. The residual channels — spin-dipole exchange with coupling m_N/f_a , loop-induced scalar admixtures, two-boson exchange, or condensate mixing — require a dedicated analysis and are not claimed here as quantitative predictions; we state only the robust null expectation for the unpolarized monopole channel.

3 Experimental Signature

The benchmark target is summarized in Table 1.

Table 1: Benchmark target of the NVG θ -DM sector, conditional on the θ -seesaw hypothesis with $c_\nu = 1$. No quantity in the right column is fitted to haloscope data.

Quantity	Predicted value	Origin
Decay constant f_a	1.07×10^{11} GeV	neutrino sector, Eq. (4)
Mass m_θ	53.2 μ eV	Eq. (2), lattice χ_{top}
Frequency f_θ	12.86 GHz	$m_\theta c^2/h$
Virial linewidth Q_a	$\sim 10^6$	halo velocity dispersion
Line width Δf_a	12.8 kHz	f_θ/Q_a
$g_{\theta\gamma\gamma}$ (KSVZ)	2.08×10^{-14} GeV $^{-1}$	Eq. (9)
$g_{\theta\gamma\gamma}$ (DFSZ)	8.10×10^{-15} GeV $^{-1}$	Eq. (9)
Coherent DM fraction	24%	misalignment, Sec. 2.6
$m_\theta f_a$ vs QCD band	−2.6% deviation	consistency check

The signal is a nearly monochromatic microwave excess at 12.86 GHz with $Q_a \sim 10^6$ intrinsic quality factor, linearly polarized along the applied magnetic field, scaling as B_0^2 , and tracking the sidereal modulation of the halo wind (a secondary discriminant).

4 Status of Published Searches

Table 2 compiles the published haloscope coverage around the predicted mass.

Table 2: Published haloscope mass windows around the NVG target. The benchmark point (53.2 μ eV) lies in the unexplored gap between the two ORGAN windows, at the lower edge of the SMASH band (50–200 μ eV) [11] that ORGAN declares as its roadmap target [16].

Experiment	Mass window (μ eV)	Frequency (GHz)	Best limit on $g_{a\gamma\gamma}$ (GeV $^{-1}$)
ADMX [14]	2.66–3.31	0.64–0.80	KSVZ reached
HAYSTAC II [15]	~ 17.3 –19.5	4.18–4.71	$\sim 10^{-13}$
ORGAN [18]	25.45–26.27	6.15–6.35	2.8×10^{-13}
NVG target	53.2	12.86	2.08×10^{-14}
ORGAN [16]	63.2–67.1	15.3–16.2	3.0×10^{-12}
ORGAN 1b [17]	107.42–111.93	26.0–27.1	cogenesis ALP excluded
MADMAX [21, 24]	40–400 (design)	10–100	prototype phase

Two remarks. First, no published limit reaches $g = 2.1 \times 10^{-14}$ GeV $^{-1}$ anywhere near 53 μ eV: the closest result (ORGAN at 25.5–26.3 μ eV) is a factor ~ 13 weaker in coupling, at a different mass. Second, the ORGAN collaboration’s stated goal is to scan the SMASH band 50–200 μ eV under a constant magnetic field [16]; the NVG benchmark sits exactly at the lower edge of that band, making a targeted single-frequency run a natural addition to their program.

5 Test Protocol

5.1 Signal-power formula and its calibration

For a microwave cavity haloscope [8, 25],

$$P_{\text{sig}} = g_{\theta\gamma\gamma}^2 \frac{\rho_\theta}{m_\theta} B_0^2 V C \min(Q_L, Q_a), \quad (12)$$

where $\rho_\theta = f_{\text{coh}} \rho_{\text{DM}}$ with $\rho_{\text{DM}} = 0.4 \text{ GeV}/\text{cm}^3$, V is the cavity volume, C the mode form factor, and Q_L the loaded quality factor.

Calibration against published data. The ORGAN collaboration quotes, for its Phase 1a apparatus, $P_{\text{sig}} \simeq 2.3 \times 10^{-25} \text{ W}$ (KSVZ) and $3.1 \times 10^{-26} \text{ W}$ (DFSZ) for a $64 \mu\text{eV}$ axion [16]. Two checks:

1. *Apparatus-independent:* the coupling-coefficient ratio $(g_{\text{KSVZ}}/g_{\text{DFSZ}})^2 = (0.97/0.36)^2 = 7.26$ versus the published power ratio $2.3/0.31 = 7.42$ — agreement at the 2% level, confirming the normalization convention of Eq. (9).
2. *Absolute:* back-solving Eq. (12) from the published $2.3 \times 10^{-25} \text{ W}$ implies an apparatus product $VCQ \approx 0.44 \text{ m}^3$ (for $B_0 = 7 \text{ T}$), consistent with an ORGAN-scale cavity at 15.6 GHz with $Q \sim 10^4$ – 10^5 . The unit chain of the present calculation is thus validated against a real haloscope at the factor-of-two level.

For the dielectric booster concept [19, 20, 21] the analogous expression is

$$P_{\text{sig}} = g_{\theta\gamma\gamma}^2 \rho_\theta B_0^2 \frac{A \beta^2}{m_\theta^2}, \quad (13)$$

where A is the disk area and $\beta^2 \equiv P_{\text{sig}}/P_0$ is the *power* boost relative to a bare magnetized mirror ($P_0 = g_{\theta\gamma\gamma}^2 \rho_\theta B_0^2 A/m_\theta^2$), following the MADMAX convention [24] in which β itself denotes the amplitude boost; all boost figures quoted in this note are power boosts β^2 . Both Eq. (12) and (13) have been re-derived within the Lorentz-reciprocity framework [25, 26]; an in-situ reciprocity measurement with a source antenna is the final normalization check of the stack before any dark-matter run.

5.2 Protocol A: resonant cavity

At $f_\theta = 12.86 \text{ GHz}$ the TM_{010} mode requires a cylinder of radius $R = x_{01}c/(2\pi f_\theta) = 8.92 \text{ mm}$. With length $L = 50 \text{ mm}$, $V = 12.5 \text{ cm}^3$, form factor $C = 0.69$, $B_0 = 10 \text{ T}$ and a superconducting (Nb) cavity with baseline $Q_L = 10^5$,

$$P_{\text{sig}} = 6.3 \times 10^{-25} \text{ W} \quad (f_{\text{coh}} = 0.24). \quad (14)$$

A single cavity reaches $\text{SNR} = 5$ only after ~ 48 years: it is a pathfinder, not the discovery configuration. The viable cavity route is a phase-locked array (Table 3); $Q_L = 10^6$ at 13 GHz is flagged optimistic (Nb BCS losses scale as ω^2), so the $Q = 10^5$ rows are the engineering baseline.

Table 3: Integration time to $\text{SNR} = 5$ for cavity arrays ($B_0 = 10 \text{ T}$, $T_{\text{sys}} = 1 \text{ K}$, $f_{\text{coh}} = 0.24$, per-cavity signal from Eq. (12)).

	$N = 64$	$N = 256$
$Q_L = 10^5$ (baseline)	276 days	69 days
$Q_L = 10^6$ (optimistic)	0.3 days	0.1 days

5.3 Protocol B: fixed-position dielectric booster stack

The dielectric haloscope [19] replaces the resonant cavity by ~ 80 dielectric disks (LaAlO_3 , $\varepsilon \approx 24$) in a strong magnetic field; constructive interference of the emitted photons provides the power boost β^2 . In a scanning experiment the disk positions must be moved mechanically — the technically hardest subsystem, now being commissioned [23]. **The point-frequency target**

eliminates this subsystem entirely: at a known frequency the inter-disk spacing is fixed, and the stack becomes a static optical element.

With $D = 1.2$ m disks ($A = 1.13$ m², the low-mass-end MADMAX aperture), $B_0 = 10$ T and $f_{\text{coh}} = 0.24$, Eq. (13) gives (Table 4). The power-boost targets $\beta^2 = 10^4$ – 10^5 are the design goals of the MADMAX program [21]; the three-disk prototype has so far demonstrated β^2 up to ~ 2000 with 13–17% uncertainty [24], which we state as the principal technical risk of this option.

Table 4: Fixed dielectric stack, $B_0 = 10$ T, $T_{\text{sys}} = 1$ K, readout bandwidth equal to the axion linewidth $\Delta f_a = 12.8$ kHz (the stack boost band $\gg \Delta f_a$).

Power boost β^2	$P_{\text{sig}} (f_{\text{coh}} = 0.24)$	$P_{\text{sig}} (f_{\text{coh}} = 1)$	$t(\text{SNR} = 5)$
10^4	3.1×10^{-23} W	1.3×10^{-22} W	18 h
10^5	3.1×10^{-22} W	1.3×10^{-21} W	11 min

Protocol B is strictly simpler than the cavity array: no cryogenic tuning mechanism, no phase-locked multi-cavity readout, a single wide-bore magnet, horn antenna and a near-quantum-limited amplifier — while the required power boost β^2 is already the stated MADMAX design target in a frequency band that includes 12.86 GHz.

5.4 Readout, statistics, and preregistered criteria

Noise. $T_{\text{sys}} \approx 1$ K is adopted: millikelvin physical temperature plus a flux-driven Josephson parametric amplifier. This is not an assumption but demonstrated technology — ORGAN’s most recent phase operates exactly this way [18].

Statistical test. With the Dicke radiometer relation

$$\text{SNR} = \frac{P_{\text{sig}}}{k_B T_{\text{sys}}} \sqrt{\frac{t}{\Delta f}}, \quad (15)$$

the *preregistered* search window combines a Monte Carlo propagation of the statistical inputs (Appendix 7), $m_3 \sim \mathcal{N}(50.3, 0.25)$ meV, $\chi_{\text{top}}^{1/4} \sim \mathcal{N}(75.5, 0.7\%)$ MeV, with explicit conditional scenarios for the unknown coefficient c_ν . We treat c_ν as a parameter, not a random error: no prior on it is motivated, so the c_ν -dependent ranges below are *sensitivity scenarios*, not statistical credible intervals.

- **Benchmark window** ($c_\nu = 1$ fixed). The target-frequency distribution has median 12.86 GHz and a 68% interval [12.67, 13.05] GHz (relative width 1.5%, dominated by χ_{top}). The preregistered window is the 3σ envelope $12.86 \text{ GHz} \times (1 \pm 4.4\%) = [12.29, 13.43]$ GHz, resolved in bins of width Δf_a .
- **Sensitivity scenario** (c_ν varied over $[1/2, 2]$). Via Eq. (6) the conditional band spans 6.4–25.7 GHz; a discovery program over this band carries the corresponding trials factor, which we state honestly.
- **Extreme envelope** ($c_\nu \in [1/3, 3]$). $f_\theta \in [4.3, 38.6]$ GHz (~ 9 times the benchmark band): the full scan cost if c_ν is never fixed by theory.

Signal power across the band. Along the curve (6), at *fixed coherent fraction*, the signal powers scale simply: $P_{\text{sig}}^{\text{stack}} \propto g^2/m_\theta^2 \propto c_\nu^0$ for the dielectric stack (exactly invariant under c_ν rescaling) and $P_{\text{sig}}^{\text{cavity}} \propto g^2/m_\theta \propto c_\nu^{-1}$ for the cavity, so the integration times of Tables 3–4 apply unchanged to the stack and rescale as c_ν^2 for the cavity at any other benchmark value of c_ν . If

instead the misalignment estimate of Sec. 2.6 is imposed, the coherent fraction itself scales as $f_{\text{coh}} \propto f_a^{7/6} \propto c_\nu^{7/6}$, which multiplies both powers accordingly. The $\pm 0.5\%$ window of the v1 draft covered only m_3 and χ_{top} and is superseded. Because the benchmark target is a point rather than a band, the look-elsewhere penalty inside the preregistered window is negligible; no trials factor beyond it is applied to the benchmark test.

Discovery criterion. A 5σ excess inside the preregistered window, persisting under independent re-taking, with the two corroborating checks: (i) power scaling $\propto B_0^2$ when the magnet current is varied; (ii) consistency of the line shape with the Maxwell–Boltzmann halo profile. Reproduction at a second apparatus with a different magnet is required before any claim.

Exclusion criterion. Exclusion is reported at two levels. (1) A null result at KSVZ-level sensitivity, $g_{\theta\gamma\gamma} \leq 2.1 \times 10^{-14} \text{ GeV}^{-1}$ across the preregistered window, excludes the *KSVZ-benchmark realization* of the θ -seesaw target at $f_{\text{coh}} = 0.24$ — and only that realization, since E/N is UV-dependent. (2) A null at DFSZ level ($8.1 \times 10^{-15} \text{ GeV}^{-1}$) additionally covers the DFSZ benchmark; excluding the neutrino–axion linkage itself would further require a fixed E/N assignment and a cosmological determination of f_{coh} . Because exclusion curves assume the full local density, the effective bound on g for a coherent fraction f_{coh} is weaker by $1/\sqrt{f_{\text{coh}}} \approx 2.0$; this rescaling must be applied when interpreting any null.

Side-channel cross-checks. The same f_a predicts $\sum m_\nu = 59 \text{ meV}$ (CMB+BAO, ongoing) and a $0\nu\beta\beta$ window $m_{\beta\beta} \in [0, 6.4] \text{ meV}$ (LEGEND-1000, nEXO): these provide framework-level corroboration independent of haloscopes.

6 Honest Caveats

1. **Framework hypothesis and its coefficient.** The θ -seesaw, Eq. (3), is a postulate of the NVG construction, and its UV origin is not derived. The coefficient c_ν — encoding the two-loop anomaly diagram, possible flavor factors and RG running — is unknown and is propagated as a parameter throughout; $c_\nu = 1$ is a benchmark, not a derived value. If Eq. (4) fails — e.g. m_3 from a future $0\nu\beta\beta$ /oscillation combination disagrees with 50.3 meV — the entire frequency target moves. The haloscope test is precisely the discriminator.
2. **Spectral premise.** The input $m_3 = 50.3 \text{ meV}$ assumes normal ordering and a near-minimal lightest neutrino mass. Oscillation experiments measure only mass-squared differences, and cosmology bounds the sum of masses without fixing m_3 at this precision. The target is therefore conditional on this premise; the Monte Carlo window of Sec. 5.4 covers the stated uncertainty, and a shift to inverted ordering or a heavier lightest state would move the target outside the benchmark window.
3. **Model dependence of the coupling.** The factor $|E/N - 1.92|$ in Eq. (9) is benchmarked to KSVZ/DFSZ; a different UV completion rescales $g_{\theta\gamma\gamma}$ without moving the frequency. The mass target is unaffected; the coupling is not predicted.
4. **Coherent fraction.** All powers use $f_{\text{coh}} = 0.24$ (misalignment only). If most of the sector’s density sits in incoherent defects, haloscope signals weaken accordingly; if defects decay into coherent modes, they strengthen. This is the dominant astrophysical uncertainty and is why the exclusion criterion carries the $1/\sqrt{f_{\text{coh}}}$ rescaling.
5. **Technology risk.** $Q_L = 10^6$ at 13 GHz and $\beta^2 = 10^5$ are program goals, not demonstrated hardware (the MADMAX prototype reached $\beta^2 \sim 2000$). The baseline rows ($Q_L = 10^5$, $\beta^2 = 10^4$) reflect the validated state of the art.

7 Conclusion

This note is a phenomenological target paper. Under the stated θ -seesaw hypothesis with benchmark coefficient $c_\nu = 1$, a single neutrino-sector input of the NVG/VMF framework fixes $f_a = 1.07 \times 10^{11}$ GeV, which through lattice-QCD physics yields a conditional point target: $m_\theta = 53.2 \mu\text{eV}$, a microwave line at 12.86 GHz with KSVZ-level photon coupling. The point lands on the QCD axion band within 2.6% (a normalization cross-check, not an independent test), sits in the currently unexplored gap between published ORGAN windows, and is directly on that collaboration’s declared SMASH roadmap. Because the target is a point, the search needs no scanning: a fixed dielectric booster stack reaches discovery-level significance in under a day of integration at design power boost $\beta^2 = 10^4$. The protocol above is preregistered, the frequency window is derived by Monte Carlo from all stated uncertainties, and the exclusion statements are scoped precisely: a KSVZ-level null closes the KSVZ-benchmark realization of the target, not every conceivable θ -model. Either outcome is informative — which is the property a target must have to deserve an experiment.

Appendix A: Reproducibility

All numbers in this note are produced by scripts in the accompanying repository (<https://github.com/infosave2007/vmf>, Zenodo-archived releases):

- `generator_project/calc_theta_haloscope_design.py` (v4) — geometry, signal powers (Eqs. (12)–(13), MADMAX β^2 convention), ORGAN calibration, integration times, published-search status, the c_ν -parametrized θ -seesaw chain, and the Monte Carlo block that produces the preregistered window of Sec. 5.4;
- `generator_project/calc_theta_haloscope_v2.py` — frequency derivation, axion-band consistency check (2.6%);
- `verification/nvg_neutrino_seesaw.py` — Eq. (4), fixing f_a ; `verification/nvg_theta_fifth_force.py` — pseudoscalar null channel.

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